BRENDA BOCHABERI

Data Scientist

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PROFESSIONAL SUMMARY

Analytically rigorous data scientist with hands-on experience building end-to-end ML pipelines and forecasting systems across time-series domains. Strong foundations in predictive modelling, data pipeline engineering, model explainability (SHAP), and MLOps deployed on AWS with Airflow orchestration, dbt transformations, and FastAPI inference. Excited to apply data science to high-stakes, fast-moving markets; actively expanding LLM/agentic workflow skills for automated data pipelines and EDA acceleration.

TECHNICAL SKILLS

Languages	Python, SQL, Bash
ML & Modelling	XGBoost, Prophet, SARIMA, Holt-Winters, Scikit-learn, time-series forecasting, walk-forward validation
Explainability	SHAP (TreeExplainer) — waterfall, beeswarm, dependence plots
MLOps	MLflow (experiment tracking, model registry), drift monitoring, model versioning
LLM / AI	OpenAI & Anthropic APIs, LangChain, agentic workflows, RAG, automated EDA pipelines
Data Engineering	Apache Airflow, dbt, Medallion Architecture (Bronze/Silver/Gold), ETL/ELT, Parquet, UPSERT/idempotency
Cloud & Infra	AWS (S3, EC2, ECS), Docker (multi-service), FastAPI, REST APIs
Visualisation	Streamlit, Matplotlib, Seaborn, PowerPoint (python-pptx)
Data Analysis	Pandas, NumPy, Jupyter Notebooks, large-scale dataset wrangling
Tools	Git, GitHub, VS Code, Linux

PROJECTS

Global Malaria Burden Forecasting Pipeline |

Python · AWS · Airflow · XGBoost · Prophet · MLflow · SHAP · FastAPI · Streamlit

- Designed an end-to-end ML pipeline on AWS medallion architecture (S3 Bronze/Silver/Gold) with Airflow orchestration on EC2 and dbt transformations directly analogous to market data pipelines.
- Trained and compared XGBoost and Prophet for multi-country time-series forecasting with walk-forward (time-aware) cross-validation; tracked all experiments in MLflow with full model registry.
- Applied SHAP (TreeExplainer) for feature-level explainability generating waterfall, beeswarm, and dependence plots to surface key drivers, supporting data-driven decision-making.
- Deployed predictions via FastAPI on ECS; built an interactive Streamlit dashboard for stakeholder consumption.

Hospital Readmission Prediction System | *Python · Docker · MLflow · FastAPI · XGBoost · SHAP*

- Built an 8-service Docker architecture integrating data ingestion, feature engineering, model training, experiment tracking (MLflow), and FastAPI inference endpoint.
- Trained an XGBoost classifier with clinically relevant evaluation metrics (AUC-ROC, precision-recall); modular design allows independent scaling of each service component.
- Applied SHAP explanations to surface per-prediction feature contributions — aligning model outputs with domain interpretability requirements.

Pharmacy Drug Sales Forecasting | *Python · SARIMA · Holt-Winters · XGBoost · Jupyter*

- Compared SARIMA, Holt-Winters, and XGBoost across 8 drug categories using walk-forward validation; recommended XGBoost for production due to its support for external regressors and non-linear patterns, with Holt-Winters as an interpretable fallback.
- Delivered comprehensive MAE/RMSE analysis with production-readiness rationale demonstrating ability to back-test and validate data-driven strategies end-to-end.

CDC Data Pipeline (Medallion Architecture) | *Python · Airflow · dbt · SQL · Parquet*

- Implemented a full medallion architecture pipeline with Airflow DAG orchestration, UPSERT-based idempotency, and data quality tests at each layer (Bronze → Silver → Gold).
- Designed star schema for analytical query performance; automated tooling streamlines downstream analyst workflows.

LLM-Integrated EDA Automation | *Python · LangChain · OpenAI API · Pandas*

- Built a LangChain agent that interprets uploaded tabular datasets and autonomously generates EDA summaries, flags anomalies, and suggests feature engineering steps — reducing manual exploration time.
- Integrated structured output parsing and tool-use patterns to handle multi-step reasoning over unstructured data descriptions.

EXPERIENCE

Data Analysis Intern — Swahilipot Hub Foundation *May 2025 – December 2025*

- Analysed large datasets to extract actionable insights that informed business decisions; collaborated with stakeholders to translate business questions into analytical projects.
- Produced data visualisations and written summaries communicating findings to non-technical audiences.

Data Scientist Intern — DataScience LTD *April 2026*

- Validated models on unseen datasets to assess generalization and real world applicability.
- Developed dashboards and analytical reports to communicate insights to stakeholders.
- Utilized SQL, Python, Pandas, NumPy, Scikit-learn, XGBoost,and power BI throughout the project lifecycle.

EDUCATION

BSc Computer Science *Expected Dec 2026*

Technical University of Mombasa

Relevant coursework: Machine Learning, Database Systems, Statistics, Data Structures, Cloud Computing
Activities: Data Science Club, AWS TUM Club, TUM Developers Club, Hackathons